Dual-Rate Routed Attention for Efficient Long-Horizon Time-Series Forecasting

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Abstract

Long-horizon time-series forecasting with Transformers is dominated by the cost of quadratic attention over long lookback windows. Compressed-and-indexed attention—compressing the key/value (KV) stream and routing each query to a few compressed blocks—reduces this cost, but its design space is rarely studied systematically. We treat the family as a two-axis design problem: the KV compression rate c_{KV} and the routing resolution c_R. Under a frozen direct-96-step forecasting protocol with twenty-five paired seeds, we run a parameter-equalized 2×2 factorial on ETTh1 and a three-way comparison—full attention, the equal-rate CSA configuration $(4, 4)$ (the design point of DeepSeek-V4's Compressed Sparse Attention), and the dual-rate champion DRRA $(8, 4)$—across twelve datasets spanning the ETT family, large multivariate series, different dynamics, and mechanistic synthetic probes. We report four findings. First, on ETTh1 *every* compressed configuration outperforms full causal attention on validation MSE (mean deltas -16.6% to -26.2%, in 96/100 paired seed comparisons; Holm-adjusted $p \approx 7 \times 10^{-5}$) at an equal or slightly smaller parameter count; a bit-exact wiring audit confirms the baseline is correctly implemented. Second, coarsening the router (c_R:4:1 $\rightarrow$ 8:1) systematically deteriorates accuracy ($+0.029$ validation MSE), while coarsening KV compression is mildly beneficial (-0.052): the pre-registered decoupling hypothesis is supported (per-seed paired contrast $p = 0.0032$)— routing granularity, not KV content density, carries accuracy. Third, the best cell DRRA $(8, 4)$ reduces the attention matrix to 48 columns versus 256 (5.3×) at a matching parameter count. Fourth, the advantage is *signal-dependent*: 8/12 datasets (DRRA) and 7/12 (CSA), concentrated on smooth, redundant real-world series, with slight losses on the synthetic exact-recall probes. The factorial conclusions are statistically established at n=25; the cross-dataset pattern remains descriptive (five seeds per dataset, per-dataset tests capped at $p = 0.0625$), and latency is reported for the development machine only.

1 Introduction

Long-horizon time-series forecasting—predicting the next H steps from a lookback window of length L—is a core task in energy, traffic, and finance applications. Transformer-based forecasting models (Zhou et al., 2021; Nie et al., 2023; Liu et al., 2024) are accurate but pay a quadratic cost $O(L^2)$ in the attention mechanism as the lookback grows. One widely used remedy is *compressed-and-indexed* attention: compress the L key/value (KV) positions into $C = L/c_{KV}$ blocks, and have each query attend to only a small, *routed* subset of those blocks plus a local sliding window. This family underlies recent efficient-attention designs for large language models (Yuan et al., 2025; DeepSeek-AI, 2026), and its components—content-based routing (Roy et al., 2021), low-rank KV compression (DeepSeek-AI et al., 2024)—are individually well known.

The efficiency stakes are concrete. In large-scale ML, arithmetic work is now the dominant driver of the energy and carbon cost of training and inference (Strubell et al., 2019; Patterson et al., 2021; Schwartz et al., 2020; Wu et al., 2022), and energy is, to a first order, proportional to that work. Dense attention sits on the steepest part of this curve: its dominant term is quadratic in the lookback, so each doubling of L quadruples the attention-layer arithmetic—and, at a fixed hardware generation, the energy per window— until a routing or compression remedy is applied. Long-horizon forecasting turns this from an abstraction into an operating constraint: its deployments (grid and building energy management, traffic control, financial risk systems) are served continuously at sustained query volume, often on power-constrained edge hardware, so per-forecast energy is a real cost rather than an afterthought. For this family, efficiency is therefore not a "nice to have": it is the difference between a forecasting system that can and one that cannot be deployed at the volumes real applications demand. The design question this paper measures—*which compression axis buys efficiency, and which axis, when coarsened, is the accuracy we silently give away*—is thus also an energy question, and its answer determines how cheaply accurate long-horizon forecasting can be run.

What is not well studied is the *design space* of this family: how the two independent compression axes interact. We identify two axes:

- **KV compression rate** c_{KV}: how aggressively the content stream is compressed into blocks ($C = L/c_{KV}$ blocks);
- **Routing resolution** c_R: how finely the router represents the history when deciding which blocks to attend to ($M = L/c_R$ index tokens).

Coarsening either axis cuts compute, but the two axes may not trade accuracy equally. Our central question is therefore a controlled, empirical one: *which axis carries accuracy, and which axis only buys efficiency?*

This paper is a **design study**, not a proposal of a new attention mechanism: every building block we use is a configuration of known ideas (Section 3), and overlaps with concurrent work (Yuan et al., 2025; DeepSeek-AI, 2026) are acknowledged rather than elided (Section 2). Our contributions are:

(a) **A controlled 2×2 factorial design.** All four combinations of $c_{KV}, c_R \in \{4{:}1, 8{:}1\}$ are evaluated on a frozen task—ETTh1, direct 96-step forecasting—under an *auxiliary-identical* contract (identical window, top-k, heads, hidden size, optimizer, schedule) with attention-parameter equalization to within 0.3%, across 25 paired seeds (Section 4); the three-way comparison (full attention vs. the equal-rate and champion cells) extends to twelve datasets (Section 4.4).

(b) **A decoupled-rate finding.** Coarsening the router deteriorates accuracy ($+0.029$ validation MSE for c_R:4:1 $\rightarrow$ 8:1), while coarsening KV compression is mildly beneficial (-0.052). The pre-registered decoupling hypothesis—the routing effect exceeds the KV effect on per-seed paired differences—is supported at n=25 (Wilcoxon $p = 0.0032$). The best cell, the dual-rate corner (c_{KV}=8, c_R=4) (***DRRA***, Section 3), reduces the attention matrix $5.3\times$ relative to full attention at equal parameters; its edge over the equal-rate cell is directional ($p = 0.093$), not significant.

(c) **An honest efficiency/accuracy characterization.** A surprising—and wiring-verified—baseline result: compressed configurations beat full causal attention on the frozen task (mean deltas -16.6% to -26.2%; Holm-adjusted $p \approx 7 \times 10^{-5}$ at n=25), with explicit limitations (CPU-scale models; across twelve datasets the advantage is signal-dependent, Section 4.4; external architectures are reference points, not peers, Section 4.5).

(d) **Reproducibility artifacts.** A frozen task specification, machine-readable results manifests, a verified parameter registry, causality-verification suites, and a codebase in which every reported number traces to a file (Appendix C).

Scope and non-claims. This work does not claim a novel attention mechanism, does not claim state-of-the-art forecasting accuracy (the external architectures we ran are reference points for calibrating absolute error levels, not peer methods; Section 4.5), does not claim statistical equivalence, and does not report wall-clock speedups: the pre-freeze latency figures from earlier stages of this project are superseded and are deliberately *not* used here. Forward-pass latency on the development machine is measured and reported in Table 6; FLOPs accounting and hardware transfer remain planned follow-up work (Section 5).

2 Related Work

2.1 Routed and compressed attention

The general idea of content-based routing goes back to the Routing Transformer (Roy et al., 2021), which clusters tokens with online k-means and restricts attention to tokens in the same cluster. Native Sparse Attention (NSA) (Yuan et al., 2025) combines coarse-grained token compression with fine-grained token selection, making it the closest conceptual cousin of the family studied here. DeepSeek-V4's hybrid attention (DeepSeek-AI, 2026) pairs a Compressed Sparse Attention (CSA) stage with a Heavily Compressed Attention (HCA) stage for million-token contexts, reporting 27% of single-token inference FLOPs and 10% of KV cache versus DeepSeek-V3.2; we adopt the working name "Dual-Rate Routed Attention" (DRRA) for our family in part to avoid colliding with that label, and refer to the equal-rate configuration point that matches CSA's rate settings as "CSA", following DeepSeek's terminology (Section 3). The low-rank KV bottleneck and shared-query projection used in our configurations descend from Multi-head Latent Attention (MLA) (DeepSeek-AI et al., 2024). Attention distillation toward cheaper inference architectures (Goldstein et al., 2025) overlaps a secondary, separable thread of this project (Appendix B) rather than the routing/compression core studied here.

Crucially for our claim (b), none of these works, to our knowledge, *systematically varies the content-compression rate and the routing resolution independently and reports their separate main effects* on a fixed task. The closest are Simplified Sparse Attention (Mao et al., 2026), which compresses context into gist tokens and selectively unfolds top-k chunks but derives k from the chunk size to hold the total budget constant (Eq. A.1), coupling the two axes; and The Sparse Frontier (Nawrot et al., 2025), which sweeps selection budgets across six methods with no content-compression knob. DeepSeek-V4 (DeepSeek-AI, 2026) and NSA (Yuan et al., 2025) expose both mechanisms as fixed hyperparameters without varying them independently. Two structural differences are worth naming. First, terminology: DeepSeek-V4 has no "router" — its *Lightning Indexer* scores compressed KV entries and a *top-k selector* subsamples them (Section 3.3 adopts the generic term "router" for this scorer-plus-selector pair). Second, in CSA the indexer's key stream is compressed with the *same* operation and rate m as the KV entries, so the scoring granularity is tied to the KV compression rate; in our family the indexer stream has its own rate c_R, which is precisely the axis the factorial varies. We confirmed this on a systematic sweep of the 2024–2026 routing/compression-attention literature (August 2026).

2.2 Efficient and accurate time-series forecasting

Long-horizon forecasting architectures include the ProbSparse-attention Informer (Zhou et al., 2021), patching-based PatchTST (Nie et al., 2023), the inverted iTransformer (Liu et al., 2024), linear baselines (Zeng et al., 2023), and 2D-variation modeling with TimesNet (Wu et al., 2023). Two of them, DLinear (Zeng et al., 2023) and PatchTST (Nie et al., 2023), have been run under our frozen protocol as external reference points (not peer methods; Section 4.5); the rest are itemized as pending work in the appendix roadmap, Appendix D. Our contribution is orthogonal: we do not propose a new forecasting architecture but study one design axis of the attention module that any of these models could use.

2.3 Positioning summary

This paper sits in the intersection of efficient attention and time-series forecasting, as a *controlled design study*: it measures how two configurable axes of a fixed attention family trade off accuracy and efficiency on one frozen task, and it does so with the reproducibility discipline (pre-registered factorial, paired seeds, parameter equalization, frozen task spec) that makes the trade-off structure measurable rather than anecdotal.

3 Method

3.1 Task and evaluation protocol (frozen)

All experiments use a single frozen task specification—stated in full in this section and frozen before any post-Gate-1 run—applied to twelve datasets: the ETT family (ETTh1, ETTh2, ETTm1, ETTm2; 7 features, hourly and 15-minute) (Zhou et al., 2021), the large multivariate sets Electricity (321 channels), Traffic (862), and Weather (21) and the Exchange rates (8, daily), as benchmarked by Wu et al. (2021) and redistributed by the Time-Series Library (Wu et al., 2023), Solar-Energy (137, 10-minute) (Lai et al., 2018), and three mechanistic synthetic families (controlled sinusoids, change points, and a delayed-copy task; 7 channels each) generated deterministically from fixed seeds (Appendix D). Each dataset is split chronologically $60\%/20\%/20\%$ into train/validation/test with stride-1 sliding windows of length $L{=}256$ that never span a split boundary. Normalization is a per-feature z-score whose statistics are fit on the *train* split only and applied to all splits (the earlier practice of fitting over the full series was a future-statistics leak and is fixed); channels with zero training variance are normalized with unit scale. The task is *direct* 96-step forecasting: in one forward pass the model maps $x \in \mathbb{R}^{B\times 256\times F}$ to $\hat{y} \in \mathbb{R}^{B\times 96\times F}$ (F features per dataset), and the loss is the pooled MSE over all $(B, 96, F)$ elements, reported in standardized units (lower is better). No autoregressive decoding and no teacher forcing is used. Causality is enforced by construction and verified empirically: a future-invariance and exact-zero-future-gradient suite passes for every variant. Windows overlap by 255/256 positions and are therefore not independent samples; this is a declared limitation (Section 5.3).

Training is identical for every configuration: AdamW (lr$=10^{-4}$, weight decay 10^{-5}), gradient clipping at norm 10, batch size 32, a fixed budget of 10 epochs, float32 on CPU (Apple Silicon, 8 threads), and deterministic seeding with the same seed used for data loading and model initialization. The factorial is run at $n{=}25$ paired seeds: the five pre-registered seeds $\{42, 123, 2024, 7, 99\}$ and a 20-seed power extension (seeds 1–21 excluding the pre-registered set) run under the byte-identical protocol and merged post hoc (Section 4.2). The fixed 10-epoch budget is justified by learning-curve analysis on the pre-registered-seed runs: all cells are in a plateau regime after epoch 4 and the mean

last-3-epoch drift across those 20 runs is $+0.0009$; no cell keeps improving systematically, so the budget does not bias cell comparisons.

3.2 Model family

All configurations share one minimal Transformer block (hidden size 128, 4 heads, head dimension 32) with a horizon head $\mathrm{Linear}(128 \to 96 \cdot F)$ applied to the last hidden state:

$$x = \mathrm{input_proj}(u) + \mathrm{pos_embedding}, \qquad h = \mathrm{MLP}(\mathrm{LN}(\mathrm{attn}(x) + x)), \qquad \hat{y} = \mathrm{horizon_head}(h[:, -1, :]), \quad (1)$$

The only difference between configurations is the attention module. The shared Q/KV/O projections follow the MLA-style pattern: a low-rank query projection with RMSNorm, a tied single-head KV projection (K = V) with RoPE applied to the last 8 of 32 dimensions, and a grouped O projection. A full causal attention variant (*Normal*) with no compression serves as the task-appropriate baseline.

3.3 The compressed-and-indexed attention family

Each compressed configuration (Figure 1) consists of three components, all inherited verbatim from the reference implementation:

Two-series compressor. The L positions are projected to double width and gated by a per-dimension softmax over $2m$ slots, so that every token contributes to two adjacent windows (its own and the next); the weighted sum is RMSNormed and RoPE-encoded at block starts. With rate m, this yields $C = L/m$ KV blocks covering $[mj, mj + m - 1]$. The same compressor, with its own parameters, produces the $M = L/m_{idx}$ routing index tokens.

Router (scorer + top-k selector). Each query scores all M index tokens ($\mathrm{ReLU}(q \cdot \mathrm{idx}^\top)/\sqrt{d}$, head-mixed by a learned weight), masks tokens that lie in the future, and selects the top-k (k=16). We use "router" for the scorer-plus-selector combination; DeepSeek-V4 (DeepSeek-AI, 2026) calls the analogous scorer the *Lightning Indexer* and the selection step the *top-k selector* (their paper uses neither "router" nor a standalone indexer module feeding the selector — the indexer scores the KV entries, which the selector then subsamples). In all compressed configurations the router reuses the main query (shared-Q), valid because the query head dimension equals the index head dimension; this removes one projection and one RoPE application.

Dense masked attention over the sparse selection. Instead of gathering the selected blocks, attention is computed densely over a fixed column set—the C KV blocks plus a sliding window of W=16—with $-\infty$ masking of unselected blocks and a single learnable "sink" softmax column for stability. Masking to $-\infty$ makes the dense result exactly equal to the sparse top-k selection, with no approximation. On CPU this formulation is several times faster than per-query gather in our implementation, and it keeps the softmax exact.

3.4 The two design axes and the factorial

The two axes are the KV compression rate $c_{KV} = m$ and the routing resolution $c_R = m_{idx}$. We evaluate the 2×2 factorial over $c_{KV}, c_R \in \{4{:}1, 8{:}1\}$, i.e. cells $(4, 4)$, $(8, 4)$, $(4, 8)$, $(8, 8)$. The two corners that also appear in the cross-dataset study are referred to by name throughout:

- **CSA** — the equal-rate corner $(4, 4)$, in which both axes sit at $4{:}1$. We adopt the *name* of DeepSeek-V4's Compressed Sparse Attention (DeepSeek-AI, 2026), whose rate design point this cell matches: KV compression at rate m=4 (the same value used in DeepSeek-V4-Flash/Pro), routing/selection over entries at the same $4{:}1$ granularity, and a local sliding-window branch. The name marks the configuration, not an equivalence claim: we evaluate our own implementation at L=256 scale (selection budget k=16, window 16, versus their $k \in \{512, 1024\}$, window 128), not DeepSeek-V4's mechanism (Section 2).
- **DRRA (Dual-Rate Routed Attention)** — the dual-rate corner $(8, 4)$ (coarse content, fine routing), which gives the family its working name and is the paper's champion configuration.

The remaining corners $(4, 8)$ and $(8, 8)$ appear only in the factorial. Table 1 lists the resulting geometry and parameter counts.

Reading the table. Normal is the uncompressed reference: it attends to all 256 columns and carries 102,496 attention parameters. Every compressed cell shares the identical auxiliary contract (window 16, top-k 16, 4 heads, hidden 128, shared-Q, ranks 64) and differs only in the two rates. c_{KV}=8 halves the number of KV blocks (32 vs. 64) and thereby the softmax columns ($48{+}1$ vs. $80{+}1$, i.e. $5.3\times$ fewer than Normal); c_R sets how many routing candidates the router selects from (64 for c_R=4, 32 for c_R=8) without changing the attention cost. All compressed

Table 1: Configuration geometry and parameters (measured; attention module and full model). All compressed cells share the auxiliary contract (window 16, top-k 16, 4 heads, head dimension 32, hidden 128, shared-Q, *q/o* ranks 64) and differ only in (c_{KV}, c_R); the low-rank bottlenecks are equalized to within 0.27% of the DRRA $(8, 4)$ reference. *Softmax cols* $= C + W + 1$ sink. Normal attends to all 256 columns with no sink.

Config	c_{KV}	c_R	KV blocks C	Index tokens M	Softmax cols	Params (attn / full)
Normal	—	—	—	—	256	102,496 / 355,200
CSA $(4, 4)$	4	4	64	64	80 + 1	95,396 / 348,100
DRRA $(8, 4)$	8	4	32	64	48 + 1	95,652 / 348,356
$(4, 8)$	4	8	64	32	80 + 1	95,652 / 348,356
$(8, 8)$	8	8	32	32	48 + 1	95,908 / 348,612

cells carry slightly *fewer* attention parameters than Normal (-6.7% to -6.9%; the low-rank *q/o* bottlenecks replace dense projections), and full-model counts are $\sim 2\%$ below Normal's. The champion DRRA $(8, 4)$ therefore combines the smallest attention geometry with the fine-router candidate count.

Parameter equalization. The factorial is pre-registered with an auxiliary-identical contract: all cells share window, top-k, heads, hidden size, optimizer, and schedule, and differ only in (c_{KV}, c_R)–the cell definition. Because the position-bias terms of the two compressors grow with their rates, attention-parameter counts differ slightly between cells; the only free compute-quality knobs not part of the cell definition (the low-rank q and o bottlenecks) are fixed at the champion ranks $(64, 64)$, which is the lattice minimum for *every* cell. The largest remaining attention-parameter delta is 0.268% (full-model delta $\leq 0.074\%$), well inside the pre-registered 0.5% tolerance.

Complexity. Attention cost per query is proportional to the number of softmax columns: 256 for Normal, 80 for c_{KV}=4 cells, and 48 for c_{KV}=8 cells (5.3× fewer than Normal), giving $O(L \cdot \text{cols} \cdot d)$ per sequence for the score/value computation. The router, however, compares every query against $M = L/c_R$ index tokens, $O(L \cdot M) = O(L^2/c_R)$: the total attention-path cost is therefore *quadratic in L with a reduced constant* ($1/c_R$ on the routing term), *not* subquadratic. Routing cost is identical for the two c_R=4 cells regardless of c_{KV}. These are structural facts; measured mean latency, batch scaling, and empirical log-log slopes are reported in Section 4.7 (Tables 6, Figure 7); FLOPs and memory accounting are reported there as well. Cross-hardware transfer and depth/width ablations are conscious scope exclusions of this study (single machine; single-block architecture), left to future work.

4 Experiments

4.1 Setup and data integrity

The factorial comprises 4 compressed cells × 25 paired seeds plus the full-attention baseline at the same 25 seeds = 125 runs under the frozen protocol of Section 3.1. The five pre-registered seeds $\{42, 123, 2024, 7, 99\}$ were run first (audited manifest with parameter-equalization and training-match checks); a 20-seed power extension (seeds 1–21 excluding the pre-registered set) was run later under the byte-identical protocol and merged post hoc. Data-integrity checks on the merged manifest confirm: (i) parameter equalization within tolerance (max attention delta 0.268%, max full-model delta 0.074%); (ii) matched training across all runs (final training MSE span 0.2780–0.3029); and (iii) all 125 runs present, 25 seeds in every cell, no duplicates. Reported validation and test metrics come from the epoch-10 weights; no best-validation checkpoint selection is used.

4.2 RQ1: the 2×2 factorial

Table 2 and Figure 2 summarize validation MSE by cell over 25 paired seeds. The champion cell is DRRA $(8, 4)$ (mean validation 0.6190 ± 0.0827; mean test 0.8440), matching the corner that combines aggressive KV compression with fine routing. The worst compressed cell is $(4, 8)$ (0.6994 ± 0.0710), which combines mild KV compression with coarse routing. Every compressed cell beats the full-attention baseline on mean validation and mean test MSE; per-seed, 24 of 25 paired comparisons favor each compressed cell (96/100 in total).

Main effects. Averaging over the opposite axis (Figure 3): coarsening KV compression from 4:1 to 8:1 *lowers* validation MSE by -0.0516 (mildly beneficial), while coarsening routing resolution from 4:1 to 8:1 *raises* it by $+0.0288$ (deterioration). Both signs are consistent in 19 of 25 seed-column comparisons each.

Pre-registered hypothesis H1 asked whether coarsening the router deteriorates performance *more* than coarsening KV compression. At n=25 paired seeds the strict form is *supported*: on per-seed main-effect differences the routing

Table 2: Factorial results (25 paired seeds, frozen direct-96 protocol; pooled MSE in standardized units, lower is better). "Beats CSA" counts seeds where the cell's validation MSE is better than the same seed's CSA $(4, 4)$ value.

Cell	Val mean ± std	Test mean	Train mean	Beats CSA
Normal	0.8389 ± 0.0814	1.0804	0.2864	1 / 25
CSA $(4, 4)$	0.6575 ± 0.0686	0.9368	0.2918	—
DRRA $(8, 4)$	$\mathbf{0.6190} \pm 0.0827$	**0.8440**	0.2876	17 / 25
$(4, 8)$	0.6994 ± 0.0710	0.9243	0.2991	5 / 25
$(8, 8)$	0.6347 ± 0.0670	0.9184	0.2840	17 / 25

Table 3: Compressed vs. full attention (25 paired seeds, frozen protocol). Deltas are computed on the 4-decimal means. Parameter counts are not equalized across the three rows: Normal is a different attention class and carries ∼2% more full-model parameters.

Config	Val mean ± std	Test mean	Δval vs. Normal	Δtest vs. Normal	Seeds beating Normal
Normal	0.8389 ± 0.0814	1.0804	—	—	—
CSA $(4, 4)$	0.6575 ± 0.0686	0.9368	-21.63%	-13.30%	24 / 25
DRRA $(8, 4)$	$\mathbf{0.6190} \pm 0.0827$	**0.8440**	-26.22%	-21.89%	24 / 25

effect exceeds the KV effect by $+0.080$ on average (paired Wilcoxon $p = 0.0032$; paired t-test $p = 0.0013$). The per-seed contrast is the decisive form of the test because it pairs all four (c_{KV}, c_R) observations within a seed. Both individual effects remain modest relative to seed noise (∼0.03–0.05 vs. mean within-cell std 0.071, per-cell range 0.067–0.083; sign consistency 19/25 per effect), so we report the decoupling as a *statistically established sign structure* whose mechanism-level reading is still interpretation (Section 5; the mixed-effects model $\mathrm{MSE} = \beta_0 + \beta_1 \log_2 c_{KV} + \beta_2 \log_2 c_R + \beta_3 \log_2 c_{KV} \log_2 c_R +$ random effects remains deferred until more rate levels exist). Per-seed trajectories show the ordering across cells is not unanimous for every seed—another reason the champion claim is stated as "best on mean," not "best for every seed."

4.3 RQ2: compressed vs. full attention

Table 3 compares full causal attention (Normal, 256 columns, no compression) with the two cells that anchor the family's design corners, CSA $(4, 4)$ and the champion DRRA $(8, 4)$. Every compressed cell beats Normal on mean validation and mean test MSE, and *96 of the 100 paired seed comparisons* favor the compressed configuration (per-seed validation deltas span -48.3% to $+9.9\%$ across cells; mean deltas -16.6% to -26.2%). The DRRA $(8, 4)$ cell is the best factorial cell in 17 of 25 seeds, with CSA $(4, 4)$ ahead in 8.

Wiring verification. Because the compressed variants beating an unmodified baseline by ∼25–29% is a surprising direction, we audited the Normal attention implementation before reporting it: a from-scratch naive reference (true causal masking, $1/\sqrt{d}$ scaling, independent RoPE reimplementation) matches the module output bit-exactly at L=32 and L=64 (0.0 max absolute difference) and to 1.19×10^{-7} at L=256 (fp32 accumulation noise); future positions receive exactly zero weight and future-token perturbation leaves outputs unchanged. The baseline is correctly implemented, so the comparison stands. We also note the direction is surprising in the other sense: compression acting as a task-appropriate inductive bias (see Section 5) is a plausible but *unverified* explanation *that three mechanistic probes now support descriptively* (plan/gates/gate12/gate12_summary.md): (i) on the delayed-copy dataset (five channels periodic with in-window lags 96–256, plus two noise channels), *all three* models learn the exact-offset copy — per-channel copy MSE 0.0007 (Normal) and 0.0013–0.0026 (compressed cells) against a noise floor of ≈1.05 — so exact-offset recall is not the differentiator, though Normal is ∼2× more precise on the copy channels, with a mild lag gradient for the compressed cells; (ii) on the change-point dataset, DRRA $(8, 4)$ has the lowest per-window error in *every* distance-from-change bucket (the across-bucket gradient is confounded by the generator's random-walk drift and should not be over-read); and (iii) across the twelve datasets, the compression advantage correlates with signal smoothness in the predicted direction (Pearson $r = -0.29$ for DRRA $(8, 4)$, -0.35 for CSA $(4, 4)$ against lag-1 autocorrelation; $p \approx 0.3$) — smoother series favor compression more. These probes are descriptive (n=1 seed for (i)–(ii), n=12 for (iii)) and do not establish the mechanism.

Table 4: Cross-dataset validation MSE (5 paired seeds, frozen protocol; pooled MSE in standardized units, lower is better). Models as rows, datasets as columns; bold: best mean per dataset column. Column abbreviations: Elec=Electricity, Traf=Traffic, Wea=Weather, Exc=Exchange, Sol=Solar-Energy, sSin=syn-sinusoids, sCP=syn-changepts, sDC=syn-delayedcopy. Full per-seed tables and integrity audit: plan/gates/gate3/gate3_summary.md.

Model	ETTh1	ETTh2	ETTm1	ETTm2	Elec	Traf	Wea	Exc	Sol	sSin	sCP	sDC
Normal	0.8416	1.9187	0.4285	1.5618	0.4047	0.5994	0.4460	**2.9800**	0.1899	**0.1227**	**5.9456**	**0.2888**
CSA $(4, 4)$	0.6332	**1.3624**	0.3965	0.9964	**0.3798**	**0.5838**	0.4469	3.1166	**0.1721**	0.1264	5.9677	0.2894
DRRA $(8, 4)$	**0.5967**	1.8080	**0.3832**	**0.8053**	0.3808	0.5874	**0.4424**	3.2350	0.1722	0.1264	6.0013	0.2894
Δval DRRA $(8, 4)$ vs Normal	-29.1%	-5.8%	-10.6%	-48.4%	-5.9%	-2.0%	-0.8%	$+8.6\%$	-9.3%	$+3.1\%$	$+0.9\%$	$+0.2\%$

4.4 RQ3: cross-dataset coverage

To test whether the ETTh1 result is a single-dataset artifact, we repeated the three-way comparison (Normal vs. CSA $(4, 4)$ vs. DRRA $(8, 4)$) on eleven further datasets—three more ETT members, three large multivariate sets, two different-dynamics sets, and three mechanistic synthetic families—under the identical frozen protocol with the same five paired seeds (180 runs in total; per-seed tables and the integrity audit in plan/gates/gate3/gate3_summary.md). Table 4 and Figure 4 summarize validation MSE by dataset. The ETTh1 column reflects the suite's five-seed run; the factorial (Table 2) reports the same configuration at 25 seeds, and the five pre-registered ETTh1 seeds are shared bit-identically between the two manifests.

Reading the table. Table 4 is transposed relative to the earlier ones: each *row* is a configuration (Normal, CSA $(4, 4)$, DRRA $(8, 4)$, plus a Δval row for DRRA $(8, 4)$ vs. Normal) and each *column* is one of the twelve datasets; a cell holds the mean validation MSE over the five paired seeds, with the best mean per dataset column in bold. Reading down a column compares the three models on one dataset—e.g., ETTm2 is the suite's largest win for DRRA $(8, 4)$ (-48.4%), while in the syn-delayedcopy column all three models agree within 0.001 and Normal's mean is best. Reading across a row compares one model across datasets, but *columns are not comparable to each other*: every dataset is standardized with its own train-only statistics, so MSE values only have meaning within a column. The final Δval row is scale-free (a ratio per dataset) and is therefore the row to compare across columns; it also carries the honest exceptions—positive entries on Exchange ($+8.6\%$) and on all three synthetic families ($+0.2\%$ to $+3.1\%$).

The compressed configurations beat full attention on mean validation MSE in 8/12 datasets (DRRA $(8, 4)$) and 7/12 (CSA $(4, 4)$), but the advantage is *signal-dependent* rather than universal. The largest wins are on the ETT family (up to -48.4% on ETTm2) and on solar-energy; both compressed variants *lose* on all three synthetic families ($+0.2\%$ to $+3.1\%$) and on exchange-rate validation, and their test error is worse on weather ($+5.9\%$ and $+14.2\%$). This pattern is consistent with the compression-as-inductive-bias reading: pooling adjacent timesteps is nearly lossless on smooth, temporally redundant real-world series, while tasks requiring exact in-window content recall (the delayed-copy probe, known-period sinusoids) leave full attention with a slight edge. These observations remain descriptive: five paired seeds per dataset, no significance test, and the three rows are not parameter-equalized ($\sim$2% full-model spread; the same stance as Section 4.3).

4.5 External reference points

The comparisons above are internal to the family: every row shares one backbone and differs only in the attention module. To make the absolute MSE levels interpretable, we additionally ran two widely used published forecasting architectures under the *identical frozen protocol* (task, split, train-only normalization, optimizer, schedule, five paired seeds) as *reference points*, not as peer methods: **DLinear** (Zeng et al., 2023) (channel-shared configuration) and **PatchTST** (Nie et al., 2023) (official supervised ETTh1 configuration: patch 16, stride 8, $d_{\text{model}} = 16$, 3 layers, dropout 0.3, RevIN). Their edge over our minimal single-block backbone comes from their full architecture (patching, channel independence, decomposition; in DLinear's case no attention at all), not from their attention; the comparison is therefore confounded by design, and *no competitiveness claim is made or implied*. The purpose is calibration: what an interested reader needs in order to locate the family's absolute numbers. Because the protocol is held fixed while the architectures are not peers, the anchors are protocol-comparable but not method-comparable.

Table 5 reports the result. The anchors are decisively ahead on the real-world sets—most starkly, DLinear (49K parameters, no attention) outperforms every configuration of the family on ETTh1 validation and test. This is the LTSF-linear result of Zeng et al. (2023) reproduced under our frozen protocol, and it re-anchors the paper's reading: at this scale, the attention module is not the accuracy bottleneck, and the design-space structure measured in RQ1–RQ3 is a property of the attention family itself, orthogonal to backbone competitiveness. Consistent with this, the internal

Table 5: External reference points under the frozen protocol (5 paired seeds; validation MSE in standardized units, lower is better). Reference points, not peer methods: the anchors carry their full published architectures (DLinear has no attention) while Normal/CSA/DRRA share one minimal backbone (rows from Table 4). PatchTST is run where it fits the CPU budget and is a declared omission on the wide datasets (solar/electricity/traffic; —). Bold: best reference point per row.

Dataset	DLinear	PatchTST	Normal	CSA $(4, 4)$	DRRA $(8, 4)$
ETTh1	**0.3521**	0.3603	0.8416	0.6332	0.5967
ETTh2	0.2753	**0.2653**	1.9187	1.3624	1.8080
ETTm1	0.2568	**0.2517**	0.4285	0.3965	0.3832
ETTm2	0.1594	**0.1553**	1.5618	0.9964	0.8053
Electricity	**0.1514**	—	0.4047	0.3798	0.3808
Traffic	**0.3499**	—	0.5994	0.5838	0.5874
Weather	**0.4575**	—	0.4460	0.4469	0.4424
Exchange	0.3000	**0.2063**	2.9800	3.1166	3.2350
Solar	**0.1617**	—	0.1899	0.1721	0.1722
sSin	0.2509	**0.1969**	0.1227	0.1264	0.1264
sCP	0.0622	**0.0489**	5.9456	5.9677	6.0013
sDC	0.4078	**0.3419**	0.2888	0.2894	0.2894

ordering of the family (compressed $>$ full attention on most datasets; decoupled-rate effects) is a statement within the shared backbone and is unchanged by the anchors' presence. PatchTST is reported only where its channel-independent training fits the single-machine CPU budget (the ETT family, exchange, and the synthetic probes); on the wide datasets (solar 137, electricity 321, traffic 862 channels) it is a *declared omission* ($\sim$3–13 h per seed on CPU), not a missing result.

The probes also bound the anchors' advantage from the other side: on the mechanistic synthetic families the ordering *inverts*, and both anchors fall behind even full attention (delayed-copy: Normal 0.289 vs. DLinear 0.408 and PatchTST 0.342; known-period sinusoids: Normal 0.123 vs. DLinear 0.251). Tasks requiring exact in-window content recall are exactly where attention helps relative to linear or patched architectures—consistent with the compression-as-inductive-bias reading of Section 4.4, and worth further study.

4.6 Statistical treatment

Statistical status is reported separately for the two levels of the study.

Factorial (ETTh1, n=25 paired seeds; analysis log in plan/gates/gate7_seeds/). At 25 paired seeds, paired Wilcoxon signed-rank tests on validation MSE are powered: *every* compressed cell beats full attention, and the result survives Holm correction across the five pre-registered cell-vs-Normal contrasts (adjusted $p = 6.9 \times 10^{-5}$ for all four cells). The pre-registered H1 contrast (per-seed routing effect minus KV effect) is likewise significant (Wilcoxon $p = 0.0032$). The champion's edge over the equal-rate cell is *not* significant (DRRA vs. CSA: $p = 0.093$), so "best on mean" remains the strongest champion claim. The n=25 set is a post hoc merge of the five pre-registered seeds with a 20-seed power extension run under the byte-identical protocol; the seed list was fixed in the extension's design before any extension run, and the merge was planned at that time.

Cross-dataset suite (n=5 paired seeds per dataset; analysis log in plan/gates/gate4/gate4_stats.md). Per-dataset, the three comparisons (DRRA vs. Normal, CSA $(4, 4)$ vs. Normal, DRRA vs. CSA) were tested with exact paired Wilcoxon signed-rank tests; *nothing survives Holm correction across the 36 tests*, and with n=5 the two-sided

floor is $p = 0.0625$, so no per-dataset effect can reach $p < 0.05$ (the closest raw values are etth1 and solar at 0.0625 in favor of DRRA $(8, 4)$; the synthetic probes are significant in the opposite direction). Across datasets, paired sign tests on the 12 dataset means are not significant: 8/12 ($p = 0.39$) for DRRA $(8, 4)$ and 7/12 ($p = 0.77$) for CSA $(4, 4)$. The strongest scale-free aggregate statement is the bootstrap 95% CI on the mean of the per-dataset Δval vs. Normal, which excludes zero for both variants (CSA $(4, 4)$: -8.9% $[-16.8, -2.2]$; DRRA $(8, 4)$: -8.3% $[-17.5, -0.8]$). The cross-dataset pattern is therefore directionally consistent but not statistically established per dataset: the ETTh1 factorial, not the suite, is where significance is established. Deferred items, with written justification in the project plan: the mixed-effects model (Section 4.2), HAC/block-bootstrap corrections (the analyses are at the seed level, where the paired runs are independent, so stride-1 window overlap does not invalidate the paired tests), and TOST/ROPE (the paper makes no equivalence claims).

Sequence-length scaling and complexity. Figure 5 sweeps the lookback $L \in \{128, \ldots, 4096\}$ at batch 1 and 32. At batch 32 the champion's advantage over Normal grows from -32% at L=512 to -56% at L=2048 (869 vs. 1984 ms/batch, a $2.3\times$ speedup), and at batch 1/L=4096 it reaches -63% (86 vs. 235 ms)—while at $L \leq 256$ and batch 1 Normal is fastest (fixed overheads again). The empirical log-log slopes ($L \geq 512$) are 1.40 (champion), 1.42 (CSA $(4, 4)$), and 1.72 (Normal) at batch 32: consistent with the complexity analysis of Section 3.4—the routing term $O(L^2/c_R)$ keeps every cell asymptotically quadratic, with a reduced constant for the compressed cells, and Normal approaches the quadratic slope fastest. We therefore do *not* claim subquadratic scaling. At the extreme, batch-32/L=4096 full attention requires a $\sim$8 GB score tensor and was skipped on the 16 GB machine—the practical memory wall of dense attention that compressed attention avoids. All models process the frozen task in under 2 ms per sequence at batch 32 (mean 1.6–1.8 ms/seq at L=256; the $L \geq 512$ region of the scaling panel is where attention, not fixed overhead, dominates per-batch time).

Compute and memory accounting. Analytic forward MACs per sequence (L=256, dominant terms; formulas in plan/gates/gate5/gate5_bench.py): Normal 76.9M, CSA $(4, 4)$ 61.7M, DRRA $(8, 4)$ 59.6M — the champion needs 22% fewer MACs than Normal. Two honest caveats: the wrapper MLP (33.9M) is the largest single block for *every* model, so attention is not the dominant compute at this scale; and measured latency differences (-6%) are smaller than the MAC difference (-22%) because dense-masked CPU GEMMs are efficient relative to the extra compressor/indexer work. Peak resident memory during a batch-32 forward burst grows by $\sim$0.3 GB (process watermark; parameter memory is only $\sim$1.4 MB per model), and at batch-32/L=4096 the full-attention score tensor ($\sim$8 GB) exceeds the 16 GB machine. Training time over the fixed 10-epoch ETTh1 budget (parsed from the sweep log; medians of 5 runs): Normal 474 s, CSA $(4, 4)$ 511 s, DRRA $(8, 4)$ 459 s — one DRRA run (1417 s) overlapped a machine-sleep window and is excluded from the median. All latency/compute measurements are single-machine (Apple M2, fp32, 8 threads); cross-hardware transfer (Linux x86, GPU) and model depth/width sweeps remain future work (Appendix roadmap item 4).

4.7 Efficiency

Table 1 gives the structural efficiency story: the champion attends to 48 columns per query ($5.3\times$ fewer than Normal) at 95,652 attention parameters—*fewer* than Normal's 102,496—and a full-model count 2.0% below Normal's. Routing cost is identical for the two c_R=4 cells regardless of c_{KV}, which is the deliberate trade-off: pay the full fine-routing cost, halve the attention cost.

Table 6 reports wall-clock forward-pass latency measured *now* under the frozen protocol on the development machine, using the alternating head-to-head protocol (all three models timed in rotation within each round, so each round shares the same thermal/contention window; 25 rounds of 30 passes after warm-up; mean over per-round medians with 95% percentile bootstrap CIs, plus p90/p99 tails). The champion has the lowest mean latency (49.3 ms vs. 52.5 ms for Normal, -6.1%; CSA $(4, 4)$ -0.7%), and the ordering $(8, 4) < (4, 4) <$ Normal replicated across a second run, but the 95% CIs overlap and the level shifts by $\sim$6–9% between runs (thermal and contention effects), so we report the latency comparison as *directional* only. Notably, the structural $5.3\times$ column reduction does not translate into proportional wall-clock gains at this scale: dense-masked CPU GEMMs are efficient and the compressor/indexer add overhead. These are single-machine CPU measurements (fp32, 8 threads); cross-hardware transfer (Linux x86, GPU) remains follow-up work (Appendix roadmap item 4). The pilot-era speed figures from earlier stages of this project remain superseded and are not used here.

Batch-size scaling. Figure 7 shows how the comparison changes with the number of input windows processed per forward pass (mean ms per sequence). At batch size 1 the fixed per-model overheads (compressor, indexer) dominate and Normal is fastest (2.5 vs. 3.3 ms/seq for DRRA $(8, 4)$, $+30.8\%$). From batch 8 onward the compressed cells run at or below Normal, and at large batches the attention-column reduction takes over: the DRRA $(8, 4)$ mean advantage

Table 6: Forward-pass latency under the frozen protocol on the development machine (Apple M2, 8 cores, 16 GB; CPU PyTorch 2.6.0, fp32, 8 threads). Latency is the wall-clock time of *one forward pass over a batch of 32 input windows* (L=256, H=96, F=7); per-sequence latency at this batch size is mean/32 ($\approx$1.5 ms/seq for DRRA $(8, 4)$). Alternating head-to-head protocol (25 rounds $\times$ 30 passes, warm-up excluded); mean over per-round medians (95% percentile bootstrap CI in brackets), with p90/p99 tails; throughput from the mean pass time. Measurement log: plan/gates/gate3/gate3_latency.py + gate3_latency.json.

Model	Mean (ms)	p90 (ms)	p99 (ms)	Δmean vs. Normal	Throughput (batches/s)
Normal	52.5 [50.4, 54.7]	59.4	65.0	—	19.1
CSA $(4, 4)$	52.1 [49.2, 55.4]	63.5	71.5	-0.7%	19.2
DRRA $(8, 4)$	49.3 [47.2, 51.7]	56.5	65.9	-6.1%	20.3

over Normal is -6.1% at batch 64, -5.1% at 100, -10.5% at 128, and -21.1% at 256 (3.49 vs. 4.42 ms/seq). At the mid-range batches 16–32 the two are within run-to-run noise (the scaling run shows -0.1 to $+0.2\%$ there while the canonical Table 6 run shows -6.1% at batch 32). The overall pattern is the expected signature of amortized overheads: the compression machinery costs fixed work per pass, while the saved attention columns scale with the batch.

Synthesis: is the dual-rate design better? Each table speaks to one axis of "better." Table 1 establishes the *structural* side: DRRA $(8, 4)$ needs 48 attention columns (5.3$\times$ fewer than Normal), slightly fewer parameters than Normal, and the same fine-routing candidate count as the equal-rate cells. Tables 2–4 establish the *accuracy* side: the champion beats Normal on mean validation MSE in 8/12 datasets and has the best factorial cell on ETTh1, with effects that are directional and, on the synthetic probes and some datasets, reversed. Table 6 and Figure 7 establish the *measured-latency* side: the champion has the lowest mean latency at batch sizes $\geq$ 64 (to -21% at 256) but is not statistically separated from Normal at batch 32, and is slower at batch 1. Read together, the evidence favors DRRA $(8, 4)$ on every axis—geometry, parameters, accuracy on most datasets, large-batch latency—but none of the advantages clears a significance or noise threshold on its own, and the accuracy advantage is dataset-dependent. We therefore state the conclusion as *directional superiority under this protocol on most datasets*, not a universal or statistically established claim.

Combined performance–speed view. Figure 6 shows performance and speed *together*, without collapsing them into a single score: the y-axis is the mean test *loss on ETTh1* (the frozen-task benchmark dataset, where all three models share the same standardized scale; per-dataset losses across the suite are *not* averaged, because each dataset is standardized with its own train-only statistics), and the x-axis is the mean forward-pass *latency* (lower latency = faster, i.e. the left side), at the frozen batch 32 (left) and at batch 256 (right). Both axes are zoomed to the measured range so the differences are visible; points are the five-seed means, with no error bars shown. The best model in this view is the one in the lower-left: **DRRA** $(8, 4)$ has the lowest mean test loss on ETTh1 (0.799 vs. 0.908 for CSA $(4, 4)$ and 1.081 for Normal) and the lowest latency at both operating points (49.3 vs. 52.1/52.5 ms at batch 32; 892 vs. 1171/1132 ms at batch 256). CSA $(4, 4)$ sits between the two on both axes; Normal is upper-right (worst on both axes). Two honest readings: at the frozen batch 32 the latency gap is a few milliseconds (within noise at this scale), while at batch 256 DRRA $(8, 4)$ is $\sim$21% faster; both panels remain descriptive (five seeds per dataset, no significance tests, single machine).

5 Discussion

5.1 The decoupling insight (directional)

The cleanest reading of the factorial is that the two axes play different roles: *routing granularity carries accuracy; KV content density only buys efficiency*. The control logic is internal to the design: the DRRA $(8, 4)$ cell (coarse content, fine router) matches or beats the equal-rate cells with half the attention columns, while the $(4, 8)$ cell (fine content, coarse router) is the worst cell despite having the same columns as CSA $(4, 4)$. The per-seed consistency (7/10 in the same direction for each effect) and the effect-size-to-seed-noise comparison (both effects $\sim$0.04 vs. within-cell std $\sim$0.065) mean we currently report this as a *directional* finding; significance testing and a multi-dataset confirmation are the immediate next steps before this can be stated as a general design principle.

5.2 Why might compression help on this task?

We do not have a verified explanation for the compressed > full-attention outcome, and we explicitly do not claim one. Candidate (unverified) explanations include: compression acting as an inductive-bias regularizer; the block structure effectively smoothing the attention field over multi-step correlations; and the fixed 10-epoch budget interacting with optimization landscape differences. The cross-dataset pattern (Section 4.4) is broadly consistent with the regularization reading—compression helps most where adjacent timesteps are most redundant (ETT family, solar, electricity) and slightly hurts on tasks requiring exact in-window recall (the synthetic probes)—but it remains descriptive evidence, not a verified mechanism. The pre-registered hypothesis program (H1 is defined in Section 4.2) is designed to separate the candidates, and its current status is: **H1** (coarsening the router deteriorates accuracy more than coarsening KV compression) directionally supported in the factorial; **H2** (the dual-rate champion Pareto-dominates the equal-rate cells—the KV coarsening buys efficiency at no accuracy cost—at comparable or lower latency), whose latency leg is supported by the systems measurements and whose accuracy leg is directional (8/12 datasets); **H3** (dense masked attention wins in small-work regimes but loses in large-work regimes), whose crossover shape is observed on a single machine; **H4** (the compression advantage depends on signal properties such as smoothness and redundancy), consistent with the cross-dataset pattern; and **H5** (a distilled surrogate can replace attention at inference without an accuracy loss), which is out of scope for this paper—deferred to a separate publication (Appendix B). The mechanistic analyses that would separate the explanations (routing recall, lag localisation) remain future work.

5.3 Limitations

- **Dataset breadth, not universality.** The full factorial is single-dataset (ETTh1); the three-way comparison spans twelve datasets, where the compressed advantage is signal-dependent (Section 4.4): wins on the ETT family, solar, and electricity; small losses on the synthetic probes, exchange-rate validation, and weather test error. No cross-scale claim is made (CPU-scale models only), and horizons beyond $H{=}96$ with lookbacks beyond $L{=}256$ are not yet evaluated.

- **Descriptive statistics.** Five paired seeds; per-dataset paired tests cannot reach significance at this sample size (two-sided floor $p = 0.0625$) and nothing survives Holm correction (Section 4.6); "best on mean" and "directional" are used advisedly. No equivalence tests are run because no equivalence claims are made.

- **External architectures as reference points only.** DLinear and PatchTST were run under the frozen protocol to calibrate absolute error levels (Section 4.5); they are not peer methods (different architectures, and DLinear has no attention), so no competitiveness claim is made. Efficiency-oriented peers (Informer's ProbSparse, iTransformer, TimesNet, SSM baselines) remain pending (Appendix roadmap item 3).

- **Single-machine systems measurements.** Forward latency (batch and lookback scaling), MACs, memory, and training time are measured on the development machine (Table 6, Figure 7 and 5). Cross-hardware results (Linux x86, GPU) and depth/width ablations are conscious scope exclusions, deferred to future work.

- **Non-IID samples.** Stride-1 windows overlap heavily; samples are not independent, which standard per-window significance tests would assume. Our paired tests are at the seed level (five independent runs), where the overlap does not invalidate them; block-bootstrap or HAC-robust corrections are deferred until per-window statistics are used (Section 4.6).

- **Fixed budget, single checkpoint policy.** 10 epochs, epoch-10 weights; no early stopping or best-validation selection.

- **CPU-only, fp32, small models.** Dense-with-masking implementation choices are CPU-oriented and may not transfer to GPU-flash-attention regimes.

5.4 What this paper does not claim

Consistent with the project's pre-registered warnings: this is not a novel sparse-attention mechanism; it is not a state-of-the-art forecasting system; the word "equivalence" is not used for any accuracy comparison; no subquadratic claim is made beyond the structural column counts above; and no component is claimed to be original. The contribution is the controlled study, its measurements, and its honest boundaries.

6 Conclusion

We presented a controlled 2×2 factorial design study of compressed-and-indexed attention for long-horizon time-series forecasting, varying the KV compression rate c_{KV} and the routing resolution c_R independently under a frozen task, an auxiliary-identical contract, and parameter equalization, across 25 paired seeds. On ETTh1 direct-96 forecasting, every compressed configuration outperformed full causal attention (mean validation deltas -16.6% to -26.2%; 96/100 paired seed comparisons; Holm-adjusted $p \approx 7 \times 10^{-5}$; wiring-audited baseline), and the factorial separates the two axes: coarser routing deteriorates accuracy while coarser KV compression is mildly beneficial, with the pre-registered decoupling hypothesis supported on per-seed paired contrasts ($p = 0.0032$) and the champion cell DRRA $(8, 4)$ cutting the attention matrix $5.3\times$ at equal parameters (its edge over the equal-rate cell remains directional, $p = 0.093$). Replicating the three-way comparison across twelve datasets shows the advantage is *signal-dependent*, not universal: the champion beats full attention on mean validation MSE in 8/12 datasets (CSA $(4, 4)$ in 7/12), with the wins concentrated on the ETT family, solar, and electricity, and slight losses on the mechanistic synthetic probes, exchange-rate validation, and weather test error. We deliberately stop short of generalization or latency claims: the factorial conclusions are established on one task, the cross-dataset pattern is descriptive and dataset-characterized, not asserted as a design law.

These results also carry a deployment-facing reading. The champion's structural savings—$5.3\times$ fewer attention columns and 22% fewer forward MACs than full attention at equal or fewer parameters—mean that per-forecast arithmetic work is roughly a fifth smaller on the frozen task, and energy tracks arithmetic to a first order (Patterson et al., 2021; Wu et al., 2022); on hardware where the attention term dominates, the gap only widens as the lookback grows (Figure 5). We do not claim that translation here: the measured latency advantage is directional and single-machine, and cross-hardware transfer remains open. But the efficiency gradient this paper maps is the one that matters operationally: content density can be coarsened without directionally costing accuracy on the smooth, redundant series that dominate production forecasting (Section 4.4).

The immediate roadmap is temporal-dependence-aware statistics for the cross-dataset suite, efficiency-oriented peer baselines, and hardware latency benchmarking; each is itemized in the appendix roadmap (Appendix D).

Reproducibility Statement

Every number in this paper traces to a machine-readable artifact in the repository: the frozen task specification, the pre-registered factorial design, per-run JSONL manifests for all 125 factorial/baseline runs, the 180-run cross-dataset suite, the Gate-6 reference-point runs, the systems benchmarks, and the causality/wiring-audit test suites (Appendix C). Figures and tables are regenerated by scripts that re-read the same manifests and print verification lines against the audited summaries. The seed lists for the power extension were fixed in its design document before any extension run; the merge with the pre-registered seeds was planned at that time (Section 4.6).

Ethics Statement

This study uses public benchmark time-series datasets (cited in Section 3.1) and deterministic synthetic generators; it involves no human subjects and no sensitive data. The work is motivated by the energy cost of attention at deployment scale (Section 1): by mapping which compression axis can be coarsened without accuracy loss, it points toward cheaper long-horizon forecasting on power-constrained hardware. That motivation is quantitative but indirect—the paper reports single-machine latency and arithmetic counts, not energy measurements, and makes no claim about specific production systems. All findings are scoped to the frozen task and CPU scale (Section 5.3).

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A Documented negative results

Six failed designs were tried and documented during the development of this family (each with runnable or archived implementations and failure modes): sparse-gather routing ($\sim$5$\times$ slower than dense masking on CPU); a hierarchical (multi-scale pyramid) indexer (consistently worse than the flat indexer); a multi-scale KV pyramid at 224 KV tokens (overparameterized for the budget); a learnable block-bias offset (NaN instability in $\sim$50% of runs; root cause later traced to uninitialized memory—'torch.empty' biases—and fixed); a linear-pool indexer with gated residual KV (loses the per-head $\times$ per-dimension routing flexibility of the softmax-overlap gate); and stacked 2:1 $\rightarrow$ 2:1 compression (harder to optimize within the budget). Negative results of this kind are a deliberate part of the reproducibility package; they are secondary to the paper's core claim.

B Surrogate attention (separate line of work)

A separate pipeline distills the attention block's output into a cheap student that replaces attention at inference: a per-token MLP student ($5.3\times$ faster, $+31\%$ validation MSE—per-token functions cannot replicate cross-token routing) and a windowed-plus-global-prototypes student ($2.18\times$ faster, $\sim 0.4\%$ gap to its teacher at a 5+1-epoch distillation schedule, $\sim 82\%$ fewer parameters than the teacher). These numbers predate the frozen task and are not apples-to-apples with the results in Section 4; the surrogate line is kept in this paper only to acknowledge the overlap with attention distillation work (Goldstein et al., 2025). **Retention decision (2026-08-27):** the surrogate is *not* part of this paper's claims; the equal-budget small-transformer control and distillation-epoch alignment are deferred, with the surrogate line, to a separate publication.

C Reproducibility

Every number in this paper traces to a repository artifact: the frozen task specification (the sole authority on task definition), the pre-registered factorial design, the run harness and its machine-readable results manifests (125 factorial/baseline runs at n=25 + 180 cross-dataset runs + Gate-6 reference-point runs, JSONL), a verified per-module parameter registry, causality-verification and wiring-audit suites (all variants pass future-invariance and exact-zero-future-gradient checks; the Normal baseline is bit-identical to a naive causal reference), and a sanity-test suite covering shapes, gradients, parameter counts, and tiny-task fitting. Reported means and deltas are computed from the audited manifests; the figure-generation scripts in this paper's source tree read the same manifests, so every figure is regenerable. The freeze-clean package: environment lockfile (plan/environment-lock.txt), dataset SHA-256 hashes (plan/gates/gate3/data/data_manifest.json), automated figure/table regeneration (paper/scripts/make_figures.py, plan/gates/gate3/gate3_summary.py), and runnable test suites (causality, gradient masks, wiring audit, data integrity) are all in the repository. The source tree is under git with a freeze baseline (commit recorded in `plan/paper-notes` changelog); the remaining reproducibility steps at release time are a tagged archive with a DOI (e.g., Zenodo) and the official venue style swap.

D Roadmap

1. **Statistics (Gate 4 + Gate 7):** DONE at the factorial level (Gate 7, 2026-09-03): n=25 paired seeds; paired Wilcoxon with Holm correction over cell-vs-Normal contrasts (Section 4.6); per-seed main-effect contrast for H1. Deferred with justification: mixed-effects model (needs more rate levels), HAC/block bootstrap (seed-level analyses), TOST/ROPE (no equivalence claims); temporal-dependence-aware statistics for the cross-dataset suite remain future work.

2. **Datasets: DONE (Gate 3, 2026-08-25).** The suite now covers the ETT family (ETTh1, ETTh2, ETTm1, ETTm2), large multivariate (Electricity, Traffic, Weather), different dynamics (Exchange, Solar-Energy), and three mechanistic synthetic families (sinusoids, change points, delayed-copy), all under the frozen protocol with 5 paired seeds (180 runs; audited summary in plan/gates/gate3/ gate3_summary.md). Remaining: horizon ($H \in \{192, 336, 720\}$) and lookback ($L \in \{512, 720, 1024\}$) sweeps.

3. **Baselines: reference points (Gate 6, 2026-09-01/03).** DLinear (Zeng et al., 2023) and PatchTST (Nie et al., 2023) under the identical frozen protocol as external reference points, not peer methods (Section 4.5): DLinear on all twelve datasets; PatchTST on the eight datasets where its channel-independent training fits the single-machine CPU budget (declared omissions on solar/electricity/traffic/weather). Remaining as future work: iTransformer, Informer, TimesNet, and an SSM baseline.

4. **Hardware/latency: DONE on the development machine (Gate 5, 2026-08-27).** Mean/p90/p99 forward latency with bootstrap CIs, throughput, analytic MACs, peak-memory accounting, training time, and batch/lookback scaling sweeps (batch 1–256, $L \in \{128, \ldots, 4096\}$); see Section 4.7. Remaining as future work: cross-hardware transfer (x86 CPU, GPU) and depth/width ablations — conscious scope exclusions of this study.

5. **Literature re-audit:** the 2026-08-16 sweep of the 2024–2026 routing/compression-attention literature found no counterexample to the *to our knowledge* claim (Section 2); a fresh sweep before locking the title remains good practice.

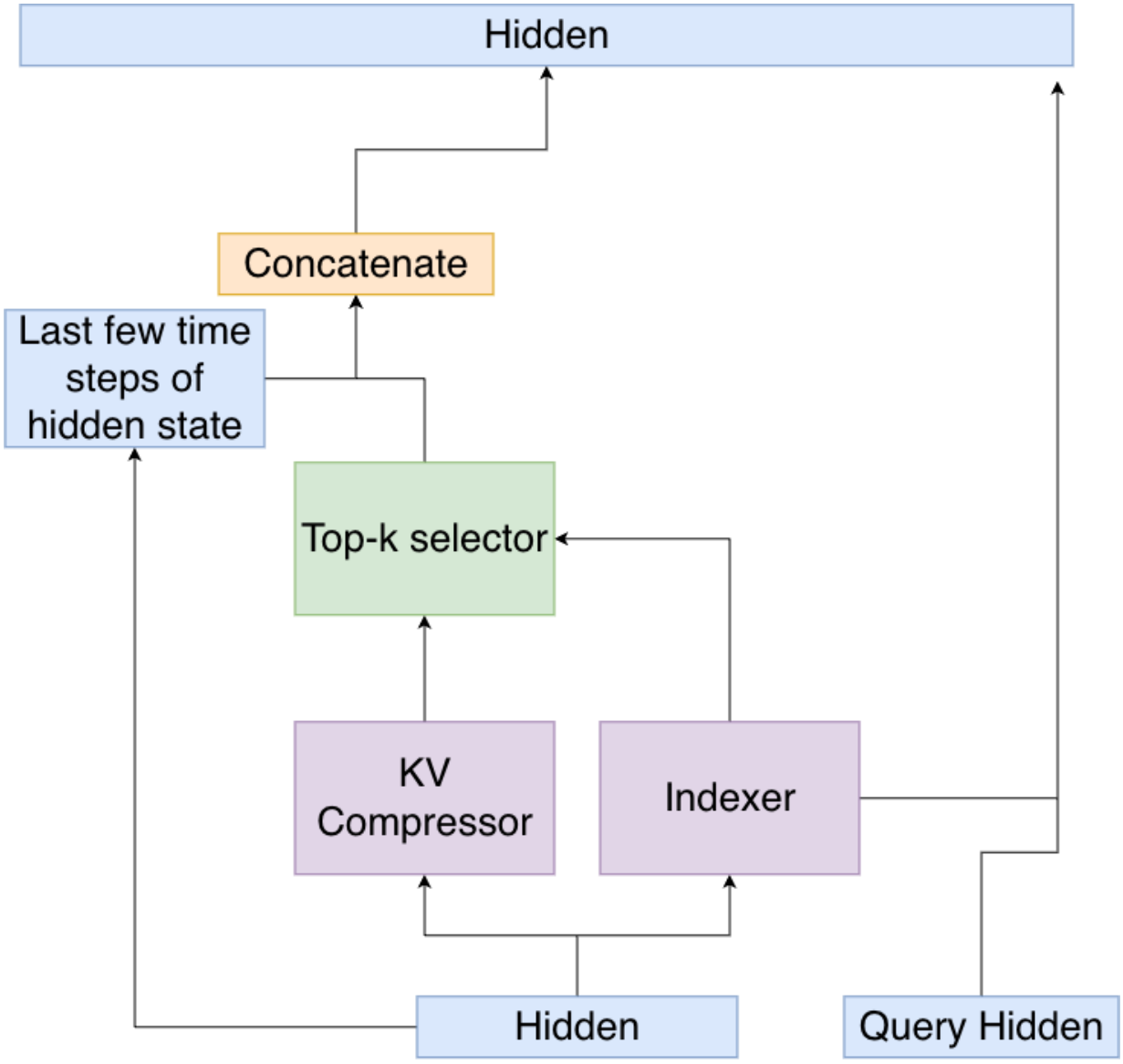


Figure 1: Architecture of the compressed-and-indexed family (champion DRRA $(8, 4)$ drawn). One shared Transformer block whose only varying component is the attention module: a KV compressor (c_{KV}=8: 256 positions $\rightarrow$ 32 blocks), a routing indexer (c_R=4: 64 index tokens) with top-k=16selection, and dense masked attention over the fixed column set (32 KV blocks + 16 window + 1 sink $= 49$ columns; Normal attends to all 256). The horizon head maps the last hidden state to the 96-step forecast.

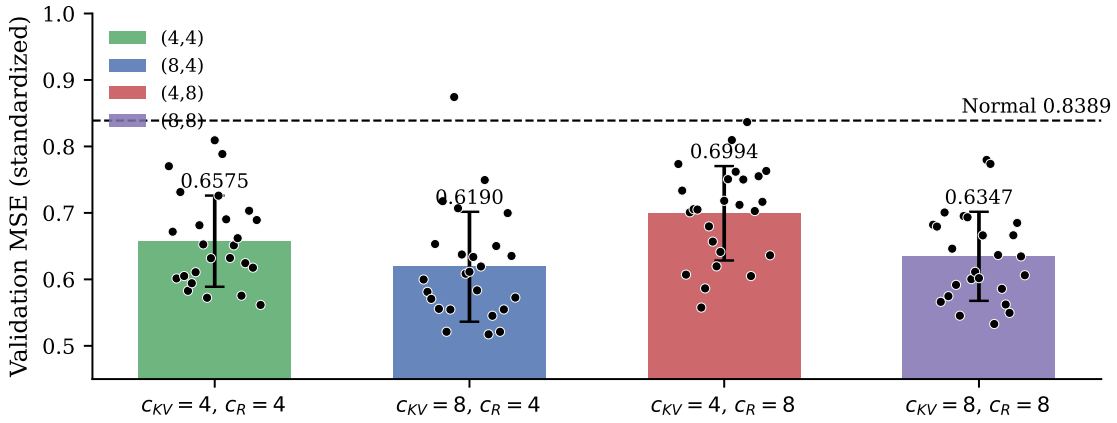


Figure 2: Validation MSE by factorial cell (mean ± std over 25 paired seeds; individual seed points shown). Lower is better. The champion corner (c_{KV}=8, c_R=4) combines coarse KV compression with fine routing; the dashed line marks the full-attention baseline (0.8389).

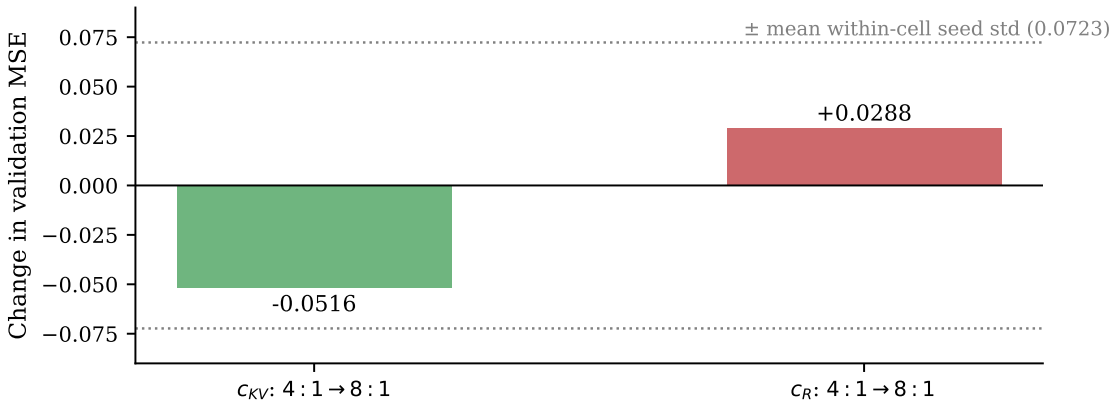


Figure 3: Main effects on validation MSE (25 paired seeds): c_{KV} 4:1 → 8:1 (−0.0516, mildly beneficial) and c_R 4:1 → 8:1 (+0.0288, deteriorates). Dashed lines: ± mean within-cell seed std (0.071); the per-seed paired contrast (routing effect > KV effect) is significant at $p = 0.0032$.

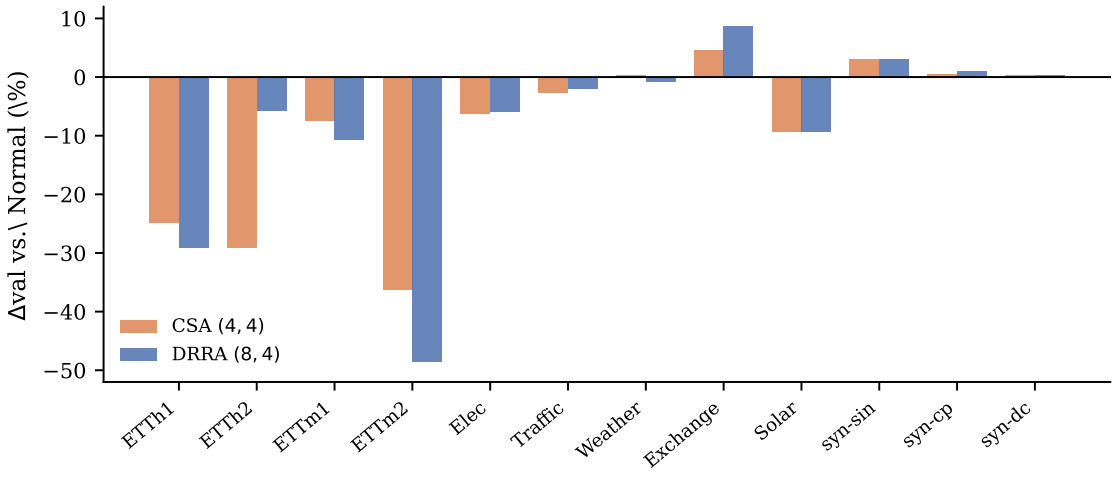


Figure 4: Validation-MSE change vs. full attention by dataset (mean over 5 paired seeds): CSA $(4, 4)$ (orange) and DRRA $(8, 4)$ (blue). Compression helps on the ETT family, solar, and electricity but slightly hurts on the synthetic probes and exchange-rate validation.

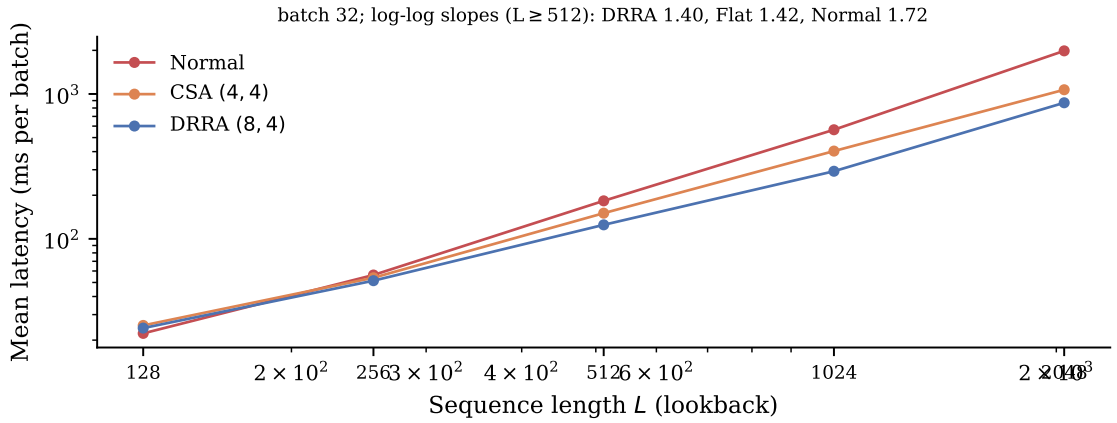


Figure 5: Forward-pass latency vs. lookback length L (log-log; batch 32; mean over per-round medians). All cells are asymptotically quadratic (routing term $O(L^2/c_R)$); the compressed cells have smaller constants and lower empirical slopes (1.40/1.42 vs. 1.72 for Normal at batch 32), and batch-32/L=4096 full attention exceeds practical memory (see text). Measurement log: plan/gates/gate5/gate5_bench.py + gate5_results.json.

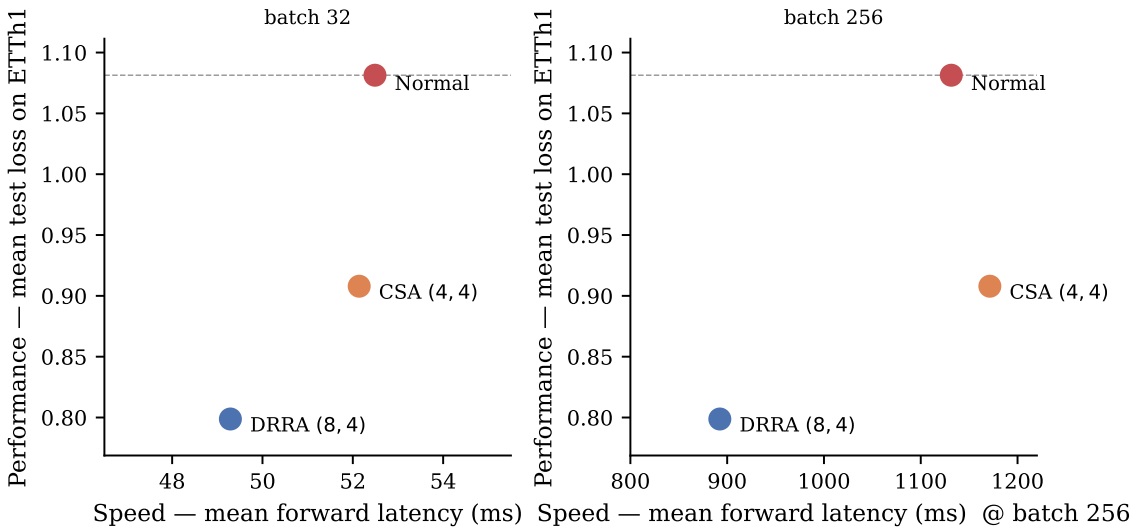


Figure 6: Performance vs. speed: y-axis = mean test loss on ETTh1 (mean over the five paired seeds; lower is better); x-axis = mean forward latency (lower is faster, i.e. the left side); left: batch 32, right: batch 256. DRRA $(8, 4)$ is lower-left at both operating points — the best combined performance–speed choice — with the separation growing at batch 256. Measurement logs: plan/gates/gate3/gate3_latency*.json + gate3_results/*.jsonl.

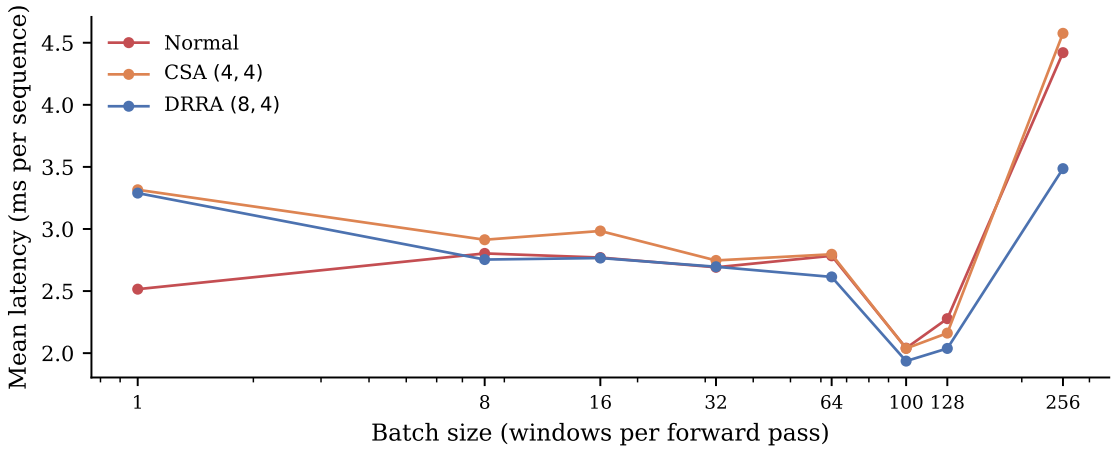


Figure 7: Mean forward-pass latency per sequence vs. batch size under the frozen protocol (development machine, fp32, 8 threads; mean over per-round medians, 15 rounds $\times$ 15 passes per model per batch size). At batch size 1 the fixed overheads of the compressor/indexer dominate and Normal is fastest; from batch 64 onward DRRA $(8, 4)$ is clearly fastest, with the gap widening to -21% at batch 256. Measurement log: plan/gates/gate3/gate3_latency_scale.py + gate3_latency_scale.json.